Лабораторная работа №7

Туйишиме Тьерри

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Файловая система Linux

Студент: Туйишиме Тьерри
Группа: НКАбд-05-25
РУДН

Цель работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов.

Команды для работы с файлами

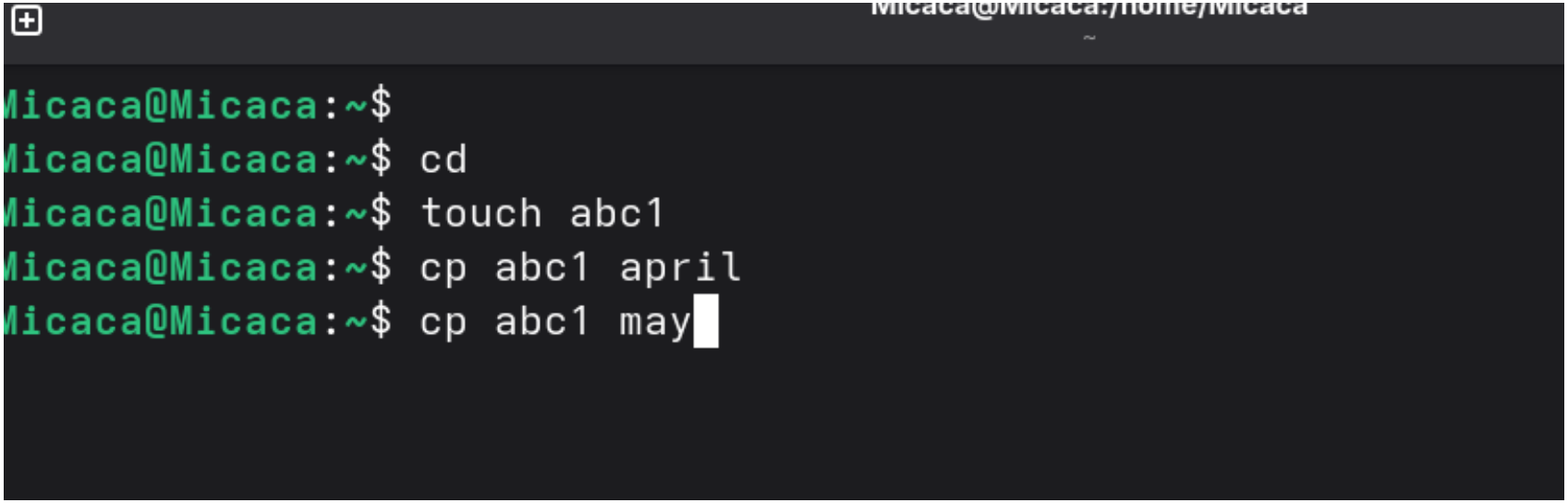

Figure 1: Создание файлов

Рис. 1: Создание файлов abc1, april и may

Копирование каталогов

```
Micaca@Micaca:~$ mkdir monthly.00
Micaca@Micaca:~$ cp -r monthly monthly.00
Micaca@Micaca:~$
```

Figure 2: Копирование каталога monthly

Рис. 4: Копирование каталога monthly

Перемещение и переименование

```
Micaca@Micaca:~$ mv april july
Micaca@Micaca:~$ mv july monthly.00
Micaca@Micaca:~$ ls monthly.00
july  monthly
Micaca@Micaca:~$
```

Figure 3: Перемещение файла july

Рис. 6: Перемещение файла july

Управление правами доступа

```
Micaca@Micaca:~$ mkdir australia
Micaca@Micaca:~$ ls -l
total 20
-rw-r--r--. 1 Micaca Micaca     0 Mar 27 16:11 abc1
drwxr-xr-x. 1 Micaca Micaca     0 Mar 27 18:26 australia
drwxr-xr-x. 1 Micaca Micaca    14 Mar 14 16:24 bin
```

Figure 4: Создание australia

Рис. 15: Создание australia

Просмотр файла /etc/passwd

```
Micaca@Micaca:~$ chmod g+w feathers
Micaca@Micaca:~$ ls -l
total 20
```

Figure 5: Команда cat

Рис. 20: Команда cat

man mount

```
Micaca@Micaca:~$ chmod u+x play
Micaca@Micaca:~$
```

Figure 6: man mount

Рис. 27: man mount

man fsck

```
MOUNT(8)                          System Administration                          MOUNT(8)

NAME
       mount - mount a filesystem

SYNOPSIS
       mount [-h|-V]

       mount [-l] [-t fstype]

       mount -a [-fFnrsvw] [-t fstype] [-O optlist]

       mount [-fnrsvw] [-o options] device|mountpoint

       mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

       mount --bind|--rbind|--move olddir newdir

       mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable]
       mountpoint

DESCRIPTION
       All files accessible in a Unix system are arranged in one big tree, the file hierarchy,
       rooted at /. These files can be spread out over several devices. The mount command serves
       to attach the filesystem found on some device to the big file tree. Conversely, the
       umount(8) command will detach it again. The filesystem is used to control how data is
```

Figure 7: man fsck

Рис. 28: man fsck

man mkfs

```
FSCK(8)                          System Administration                          FSCK(8)

NAME
       fsck - check and repair a Linux filesystem

SYNOPSIS
       fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--]
       [fs-specific-options]

DESCRIPTION
       fsck is used to check and optionally repair one or more Linux filesystems. filesystem can
       be a device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a
       filesystem label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or
       LABEL=root). Normally, the fsck program will try to handle filesystems on different
       physical disk drives in parallel to reduce the total amount of time needed to check all
       of them.

       If no filesystems are specified on the command line, and the -A option is not specified,
       fsck will default to checking filesystems in /etc/fstab serially. This is equivalent to
       the -As options.

       The exit status returned by fsck is the sum of the following conditions:
```

Figure 8: man mkfs

Рис. 29: man mkfs

man kill

```
MKFS(8)                          System Administration                          MKFS(8)

NAME
       mkfs - build a Linux filesystem

SYNOPSIS
       mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
       This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

       mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The
       device argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file
       that shall contain the filesystem. The size argument is the number of blocks to be used
       for the filesystem.

       The exit status returned by mkfs is 0 on success and 1 on failure.
```

Figure 9: man kill

Рис. 30: man kill

Выводы

Приобретены практические навыки работы с командами для файлов и каталогов, управления процессами и обслуживания файловой системы.

Спасибо за внимание!